

Dear Dr. de Ruijter-Villani and Dr. Claes,

My name is Varvara Lazarenko, and I would like to express my strong interest in the PhD position in Aneuploidy in Mammalian Embryos / Clinical Veterinary Embryologist. This inspiring position perfectly matches my long-standing interest in biology and my passion for animal sciences. Therefore, I believe I can bring real value to the project through both my education and hands-on experience, which have prepared me well to study the challenges of translational embryology and reproductive genetics.

I hold a BSc in Biology and an MSc in Medical Biology, where I gained a solid biomedical foundation and critical insight into medical technologies and biomedical challenges. During my studies, I built a broad set of wet lab skills, including PCR and qPCR, gel electrophoresis, RNA and DNA extraction, enzyme-linked immunosorbent assay (ELISA), cell and tissue culture, and immunohistochemistry.

During one of my BSc research projects, I studied the potassium channels of fish cardiomyocytes via the whole-cell patch clamp technique. For my BSc thesis and later as a Junior Research Assistant in the same laboratory, I studied the role of acid-sensitive two-pore potassium channels (TASK-1) in the vascular tone regulation in rat arteries using the wire myography, RNA extraction, and qPCR. Both patch clamp and wire myography are techniques that require fine motor skills and careful handling under the microscope, experience that I believe translates well to embryologic micromanipulation such as IVF.

I am a team player and fast learner; during my MSc internship at the Donders Institute, I quickly learned to operate a complex two-photon calcium imaging setup and was conducting *in vivo* experiments investigating social fear learning in mice. This experience enhanced my adaptability, precision, and capacity to handle technically demanding systems in a fast-paced environment. My time as the Junior Research Assistant also taught me how to collaborate across interdisciplinary teams and communicate results effectively through both publication and presentation.

Discovering this exciting PhD opportunity felt like a revelation because it brings together my passion for animal research and my ambition to contribute to translational science. I am especially enthusiastic about exploring post-zygotic aneuploidy using equine embryos due to the clear relevance to both veterinary and human reproductive medicine, as well as contributing to the development of non-invasive PGT systems for embryo screening. The integration of clinical activities and aneuploidy research, along with access to state-of-the-art imaging and sequencing technologies, makes this project highly appealing to me, and I would love to be a part of it.

I am proficient in written and spoken English, have basic French and German skills, and am actively improving my Dutch knowledge. I thrive when helping others and strongly value knowledge sharing. With experience mentoring first-year students and supporting them in understanding core biological concepts, I am very excited about the opportunity to take part in teaching during this PhD. I also look forward to being part of the Utrecht Graduate School of Life Sciences, which I see as an excellent platform for both academic and professional development.

To sum up, I hope that my interdisciplinary training, practical lab experience, and dedication to research and education make me a worthy candidate for this position. I would be honoured to join Utrecht University and contribute meaningfully to this project.

Thank you for considering my application. Should any questions arise, please do not hesitate to contact me.

Warm regards,
Varvara Lazarenko